LYNX Architecture and Setup Documentation

An AI decision-support system for NSE equities: PRISM encoders and the VANTA decision pipeline. Final architecture, version 6.0

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Document control

Field	Value
Document	LYNX Architecture and Setup Documentation, v6.0 (final architecture)
Date	20 October 2026
Purpose	One self-contained description of the whole project: the idea, the data, the models, the decision layer, the platform, the setup, the tests, the build plan and the risks. Written for supervisor review and as the working reference for the build.
Supersedes	v5.0 (9 October 2026), and through it the v3 plan, the v4 addendum, the v4 consolidated plan and the VANTA update note. Those remain as history only.
Changes from v5.0	VANTA is defined as a pipeline of four named components (NEL, VN-ListNet, RW-CS, FCC), each tied to a test and an ablation. New PRISM compute component WFP (warm-started walk-forward pretraining) to bring per-fold pretraining inside the free GPU budget. New contributions statement (Section 1.1), invariant N14, ablations A17 and A18, a validation-chosen weight decay τ, and updated build checks, risks, open decisions, glossary and references. The proposed “gate-first inference” idea was assessed and rejected (Appendix B).
Team	Aaron Abraham Mathew, Abel Tijo, Anitta Stalen, Christapher Thomas Rajesh (architecture and development lead)
Supervisor	Dr Renisha P Salim
Predecessor	SignalForge, a production US equity signal system. Its dataset is reused here as an unlabelled pretraining corpus.
Status of the evidence	The system is partly built (Section 15). **No result in this document comes from the full model on real NSE data.** The only performance figures are from synthetic simulation and are design evidence, not findings.

HOW TO READ THIS DOCUMENT

Sections 1 to 3 explain what LYNX is and why. Sections 4 to 9 are the architecture, from data to the published signal. Sections 10 to 12 explain how the system is trained, judged and kept honest. Sections 13 and 14 cover delivery and setup. Sections 15 to 17 cover the build plan, the risks and how the team works. Appendices hold the data contracts, the decision log and the reference list.

1. Executive summary

LYNX ranks the roughly 700 most liquid stocks on India's National Stock Exchange (NSE) and publishes a short list of trade ideas, each with an entry, a stop-loss and a target, plus a plain-language explanation. It never places trades. It is built as a final-year capstone and is to be delivered as a web app and as an installable mobile app on the App Store and the Play Store.

The project has three ideas that set it apart from a standard "predict the stock" model.

1. **It judges itself on money after costs, not on accuracy.** Every training label comes from simulating the exact trade that would have been published: next-day open entry, no fill if the stock opens locked at its upper circuit, an ATR-based stop and target, delayed exits when a stock is locked at its lower circuit, and the real Indian cost stack (STT, brokerage, GST, stamp duty, exchange and SEBI fees, and impact cost) at the rates in force on that date.
2. **It publishes only what it can defend.** A conformal selection gate compares each day's candidates against a year of matured, out-of-sample outcomes and publishes a name only if the evidence says it is unlikely to lose after cost. Zero names on a bad day is a correct answer. The gate targets a stated share of losing signals (30%) and the system reports how well it actually hit that target, by market regime.
3. **It is built to catch its own silent failures.** Most quantitative projects fail by leaking the future into the past without any error being raised. LYNX attaches an adversarial test to every risky component: a hand-built input with a known answer, checked to fail against a deliberately broken implementation.

The model has two named parts. **PRISM** is the representation: a patch transformer pretrained on price sequences from two markets, plus a two-layer relational graph network over six kinds of stock-to-stock link, including promoter-group membership, which has no US equivalent. **VANTA** is the decision pipeline on top. It is a named pipeline, not a new algorithm: four components, each built on published methods and each tested and ablated on its own (Section 9.0).

A language layer (a small open LLM, Qwen3-4B, fine-tuned with LoRA) reads announcements and news and writes structured, verifiable facts to a knowledge graph. It is extractive only. **The LLM never decides anything**; it feeds the graph and it narrates decisions that were already made.

> **WHAT IS NEW, AND WHAT IS NOT**
>
> Every component of VANTA has prior art: triple-barrier style trade labelling (López de Prado), ListNet ranking (Cao et al.), and conformal selection with false-discovery-rate control (Jin and Candès; Barber et al.). The contribution claimed is the **combination**, applied to NSE rules and evaluated net of Indian costs. In the wording used throughout this document, it is a combination not reported in the literature we reviewed. LYNX does not claim a new algorithm, and does not claim to be first.

1.1 Contributions, stated precisely

VANTA is the name of the LYNX decision pipeline. It is not the name of a new algorithm. The components below keep their original names and citations. What LYNX contributes is the specific adaptations, the way they are combined, and the measurement of each one.

Code	Component	Built on	What LYNX adds	Claim level	Measured by
NEL	Net Executable Labels	Triple-barrier labelling (López de Prado 2018)	Labels from a simulated NSE trade: circuit-lock no-fill and delayed-exit rules, ATR stop and target, and the date-versioned Indian cost stack	**Primary**	A2, N1 to N4, N6
RW-CS	Regime-Weighted Conformal Selection	Conformal selection (Jin and Candès 2023a), weighted conformal p-values (Barber et al. 2023; Jin and Candès 2023b)	Calibration weights from recency and the **filtered** HMM regime posterior, a Kish effective-n fallback, and a hard cap of 10	**Primary**	A10, A16, N5, N7, N13
VN-ListNet	Volatility-Normalised ListNet with soft regime gating	ListNet (Cao et al. 2007)	Gain divided by ATR% · √h, winsorised on the training fold, and heads blended by the filtered posterior	Supporting	A3, A9, N10
FCC	Full-Candidate Calibration	Standard conformal practice	Calibrates on every scored, matured candidate, never on published names only	Design rule (not a novelty claim)	N8, A17
WFP	Warm-started Walk-forward Pretraining (PRISM, not VANTA)	Continued pretraining (Gururangan et al. 2020)	Each fold's encoder continues from the previous fold's checkpoint, so per-fold pretraining stays leak-free while costing a fraction of training from scratch	Supporting (compute)	A18, N14

Position against the closest prior work found. Kaya and Nguyen (2025) apply standard conformal prediction sets to US stock selection, with a random train-test split, no transaction

costs and no false-discovery control. Kato (2024) uses conformal prediction intervals to choose portfolios, with no false-discovery control and no regime weighting. Jin and Candès developed weighted conformal selection for applications such as drug-candidate screening, not equities. Regime-weighted conformal selection over cost-aware executable labels on NSE equities was not found in that search. The search was targeted, not a systematic review, so the claim is worded “to our knowledge”.

Suggested wording for reports and slides: *“VANTA is the decision-layer pipeline of LYNX. It combines cost-aware executable labelling, volatility-normalised listwise ranking and regime-weighted conformal selection with false-discovery-rate control. To our knowledge this combination has not been applied to Indian equity signals, and each component’s contribution is measured by ablation.”*

What is expected to hold, stated in advance. NEL and FCC are the components most likely to matter. RW-CS is the real research bet and could show only a modest gain over age-only weights. VN-ListNet is the component most likely to tie or lose to the CatBoost baseline B1. Each outcome is reported as found.

2. Problem, goals and constraints

2.1 The problem

A retail investor in India faces roughly 700 liquid stocks and a flood of noisy advice. Most published “AI stock predictors” fail in one of four ways: they report accuracy instead of profit, they ignore costs and market rules, they leak future information into their tests, or they publish a call every day whatever the evidence. LYNX is designed against each of these.

2.2 Goals

#	Goal	How it is measured
G1	Rank the investable NSE universe by expected net return over 5, 20 and 60 trading sessions	Rank IC, net return per published signal, NDCG@10
G2	Publish only signals the evidence supports, with the loss share controlled	Realised false-discovery rate (FDR) against the 30% target, by fold and regime
G3	Beat honest baselines net of Indian costs, or report plainly that it did not	Ablations A0 to A16 and baseline B1 (CatBoost), with seed and fold spread
G4	Show where each component helps	One-claim-per-ablation design (Section 11)
G5	Ship a web client and a store-published mobile client on one verified backend	Stage 26 acceptance checks
G6	Be reproducible: any published signal can be traced to the model, labels, costs and calibration rows that produced it	Version tags, MLflow registry, immutable logs

2.3 Non-goals

- LYNX does not execute trades, hold funds, or connect to a brokerage account for orders.
- LYNX does not give personalised advice. Every signal is general information and carries a disclaimer.
- LYNX does not claim accuracy figures. Section 11.4 states in advance what results would be believable and what would indicate a bug.
- VANTA v1 does not cover intraday trading. See Section 2.5.

2.4 Constraints that shape the design

Constraint	Consequence
About 5,000 effective supervised samples (one per trading day, not per stock-day, because the 700 stocks share one market factor on any day)	Keep the supervised head small. Spend compute on pretraining and on ensembling, never on a bigger task head or architecture search.

Constraint	Consequence
Local GPU is a 4 GB RTX 3050 Ti	Pretraining and LLM fine-tuning run on cloud GPUs (Kaggle T4 or Colab). Supervised fine-tuning and nightly inference fit locally. NLP inference and training are never run on the card together.
Free data only	NSE archives, GDELT through BigQuery, NSE/BSE announcements, RBI and NSDL. Parsers must survive format changes, for example the July 2024 bhavcopy cutover.
Team of four, with AI coding tools used heavily	Correctness comes from targeted invariant tests, not from code review alone. A `CLAUDE.md` file carries the forbidden list into every coding session.
Paid tiers, and “entry, target, stop-loss” are the textbook form of a recommendation	SEBI Research Analyst registration may apply once money is involved. Payments are built in test mode, and the question must be resolved before either store submission.
Supervisor requirement: the app must ship on both stores	Mobile is mandatory, not a stretch goal. Backend first, clients second.

2.5 Scope decision flagged for confirmation: horizons

Earlier documents described three horizons: intraday, swing and long-term. VANTA v1 simulates trades on **daily bars** and covers horizons of **5, 20 and 60 sessions**. Intraday trades need path simulation on 5-minute bars, which is deferred. This is consistent with the descope order in Section 15.4, where intraday is the first item to go, but it is a change from the earlier product description and should be confirmed by the team and the supervisor.

3. Users, tiers and what is published

3.1 What a user sees

For each published name: the stock, the horizon, an entry rule (next session's open), a stop-loss and a target (both as fixed multiples of the stock's 14-day ATR), a short explanation grounded in the model's attribution and retrieved evidence, a timestamp, and a disclaimer. Every signal is written to an immutable log so that the public track record can be audited.

3.2 Tiers

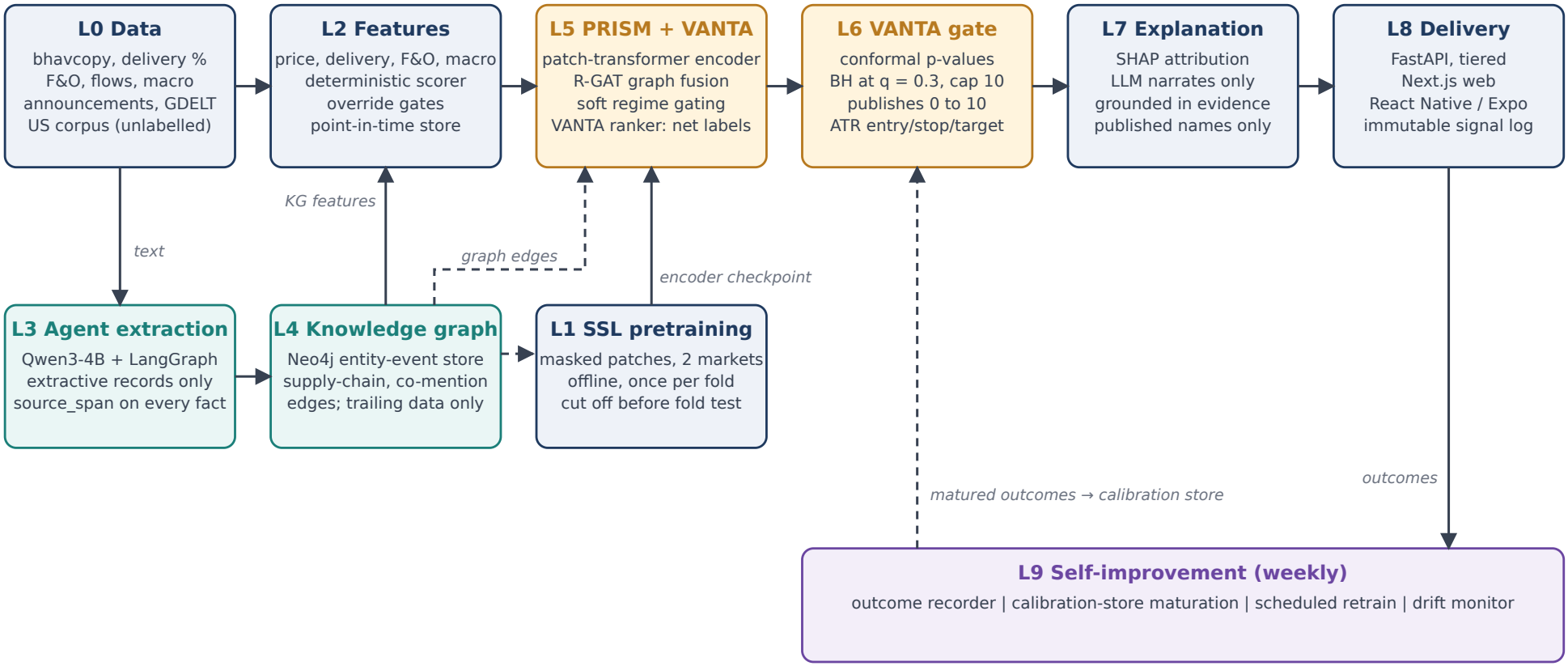
Tier	Content	Notes
Lite	A limited view served from its own endpoint family	Exact contents are fixed at Stage 26. A Lite token calling a Pro endpoint must receive a 403 with no body.
Pro	The gated VANTA list: names that passed the selection gate, with entry, stop and target	
Max	The gated list, plus the ungated ranking shown as a **watchlist without entry, stop or target**, clearly marked as not passing the gate	Showing trade levels for names the gate rejected would contradict the gate's own promise and would add exposure on the SEBI question.

4. System architecture

4.1 Nine layers

L	Layer	Cadence	Responsibility
0	Data	Nightly and bulk	Ingest, validate, adjust for corporate actions, enforce point-in-time correctness
1	SSL pretraining	Per fold, offline	Masked patch reconstruction over the two-market corpus. Produces the encoder checkpoint.
2	Features and scorer	Nightly	Price, delivery, F&O, flows and macro feature blocks. Deterministic scorer kept as a feature and a fallback. Override gates applied.
3	Agent extraction	Nightly	Qwen3-4B with LangGraph over announcements and news. Structured, extractive records only.
4	Knowledge graph	Nightly write, monthly graph rebuild	Neo4j entity-event store. Emits relational features and graph edges.
5	PRISM encoders and VANTA ranker	Nightly	**The only decision-maker.** Scores every candidate for each horizon.
6	VANTA gate (RW-CS)	Nightly	Calibration, selection, publish set, ATR trade levels
7	Explanation	Nightly, published names only	LLM narration grounded in SHAP attribution and retrieved evidence
8	Delivery	Continuous	API, web and mobile clients, tiers, immutable signal log
9	Self-improvement	Weekly	Outcome recording, calibration-store maturation, scheduled retraining, drift monitoring

LYNX system: nine layers, one decision path



The nine layers and the single decision path

Layers 1, 3 and 4 are new relative to SignalForge. Everything above Layer 5 supplies features. Everything below Layer 6 narrates or delivers. Nothing outside Layers 5 and 6 can add, remove or reorder a published signal.

4.2 The decision path, end to end

Data and features (L0, L2) feed the encoder and graph (L5). Text becomes graph facts through the extraction layer (L3, L4), and those facts return as features and graph edges. The encoder, pretrained offline (L1), and the regime posterior feed the VANTA ranker. The ranker's scores are the only input to the gate (L6). The gate decides what is published. The explanation layer (L7) is written afterwards, only for published names. Delivery (L8) serves the result, and the self-improvement loop (L9) records what happened and feeds matured outcomes back into the calibration store.

4.3 The daily cycle

All times are IST. The nightly run follows the bhavcopy publication, and the morning run only refreshes and re-filters.

Time	Step
18:00	Bhavcopy, F&O, delivery and ban list land
18:30	Ingest and validate, with row counts checked against the expected universe
18:45	Corporate action adjustment. This runs before any feature computation.
19:00	Feature blocks written to the point-in-time store
19:15	Agent extraction over new announcements and the GDELT delta into the knowledge graph
19:30	Knowledge-graph feature emission. Graph rebuild on the first session of each month only.
19:45	Encoder and ranker inference for all horizons
20:00	Filtered regime posterior, calibration lookup, p-values, BH selection, publish set
20:15	SHAP attribution and LLM explanation, published names only
20:30	Snapshot to Postgres and Redis, immutable log write, MLflow record
08:00	Publish swing and long-horizon signals
09:00	Pre-open re-filter: drop names at an upper circuit or in the F&O ban list

Outcome recording runs weekly and as each horizon matures.

4.4 What stays deterministic

The original SignalForge scorer (a weighted score over four rule layers) is kept as an auditable, cheap fallback and as a feature provider. One change was needed. Its rules were absolute, for example "RSI below 35 adds 20 points", which fires for every stock in a market-wide sell-off. Its

outputs are therefore **cross-sectionally z-scored within each date** before they enter the feature matrix. A side benefit is robustness to level shifts: when peak-margin rules changed in 2021, every stock's delivery percentage moved together, and a per-date normalised feature saw no discontinuity.

> **THE TWO-PART DEFENCE AGAINST NON-STATIONARITY**
>
> Cross-sectional normalisation removes level breaks. Rolling training windows remove relationship breaks. Neither alone is enough. Together they answer Indian market structure changes such as demonetisation, peak margin, mutual-fund recategorisation, T+1 settlement and F&O curbs.

5. Data architecture

5.1 Sources

Need	Source	Notes
Daily OHLCV and delivery %	NSE bhavcopy and `sec_bhavdata_full` (scraped)	The backbone. Delivery percentage exists nowhere else and is the strongest India-specific feature. yfinance is a cross-check only, because it drops delivery data and its corporate-action adjustment for Indian tickers is unreliable.
F&O data	NSE F&O bhavcopy	Open interest, rollover, ban list
Circuit data	NSE price-band hit lists	Drives lock detection in the label simulator
Impact cost	NSE published per-security figures	Feeds the liquidity screen and the cost model. A flat slippage assumption is not used.
Indices	Nifty, Bank Nifty, India VIX, Sensex	Display and regime features
Flows	NSDL and NSE	FII and DII daily net flows
Macro	RBI DBIE, MOSPI	Repo rate, CPI, IIP, USDINR
Historical constituents	NSE index-change announcements	Assembled by hand. Tedious, and a real differentiator.
Announcements	NSE and BSE feeds	Highest signal-to-noise text source: official, structured, timestamped and ticker-tagged by construction. Built before the news pipeline.
News history	GDELT 2.0 GKG through BigQuery	Free historical news at scale. Entity resolution to NSE tickers is built, not assumed. RSS feeds have no history and serve as a live layer only.
US pretraining corpus	SignalForge dataset (~800 tickers, 20 years)	Unlabelled use only
Intraday bars (deferred)	Kite Connect, Dhan or SmartAPI	5-minute bars, not 1-minute

5.2 What was measured, not assumed

These facts come from probing NSE's live archive during Stage 3. They set the effective window of the whole project.

Dataset	First available	Coverage
OHLCV prices (bhavcopy)	3 November 1994	About 31 years
Delivery % (`sec_bhavdata_full`)	30 September 2019	About 6 years
India VIX on disk	14 May 2014	Regime features start here

Consequences:

- **Walk-forward depth.** Prices support up to roughly 25 yearly test folds under the 4-year train, 1-year validate, 1-year test scheme. Features that need later data, such as delivery

and VIX-based regime inputs, cover fewer folds.

- **Delivery is a per-fold optional feature.** Folds testing before September 2019 mask it with a missingness indicator. The difference between delivery-enabled and delivery-masked folds becomes ablation evidence.
- **Universe growth is real.** File size grows about 37 times from 1994 to 2024 because NSE listed a handful of securities then and thousands now. Early folds have a smaller investable universe. This is reported, not corrected.
- **Deep history is an evaluation benefit, not a training benefit.** 1995 data trains the 1999 model and never touches the 2024 model.

5.3 Format and parser rules

- **July 2024 cutover.** NSE circular 62424 replaced the legacy 14-column bhavcopy with the 34-column CM-UDiFF format from 8 July 2024, and the column names are entirely different. Ingestion branches on this date. Equity rows in the new file are filtered by instrument type.
- `sec_bhavdata_full` **has one stable format across the cutover** and is the preferred source from 30 September 2019, because it gives price and delivery together.
- **Quirks that must be handled:** every column name except the first has a leading space; a literal `-` means *missing*, not zero; dates are `03-Jun-2024` and are parsed with an explicit format.
- **Never zero-fill a missing feature.** A stock with zero delivery is a different fact from a stock with no delivery data. Missing becomes NaN and carries a missingness indicator. A zero-filled RSI would read as “maximally oversold”, which is a lie.
- **404 is a normal answer.** Weekends, holidays and dates before a dataset starts all return 404. Ingestion treats it as “no trading day”.
- **Politeness.** A browser-like User-Agent is required. Requests are sequential with a delay of at least 0.4 to 0.5 seconds. Retries use exponential backoff for intermittent DNS failures.

5.4 Storage and ingestion design

```
data/raw/
  bhavcopy/        one Parquet file per trading day (price)
  delivery/        one Parquet file per trading day (sec_bhavdata_full)
  fno_bhavcopy/    F&O files
  announcements/   NSE and BSE announcements
data/processed/
  corporate_actions.parquet, pit_features.parquet, *_checkpoint.json
```

Parquet rather than CSV, because training reads thousands of these files repeatedly. One file per day, because it makes ingestion resumable, makes point-in-time reads trivial, and limits a corrupted download to one day. All writes are atomic: data is written to a temporary file and renamed, so an interrupted run can never leave a truncated file that reads as valid. A manifest records, for every date, whether it was fetched, was a holiday, failed or was empty, so a question like “why does 2011 look odd” has an answer months later. Re-running ingestion skips completed dates.

5.5 Corporate actions, universe and point-in-time rules

- **Corporate action engine (Stage 4).** Prices are adjusted for splits, bonuses and similar events. The check: no adjusted daily return exceeds its price band without a matching action record, across the full history. Lock detection always uses *raw* prices against the band, and returns use *adjusted* prices.
- **Universe (Stage 5).** Per-year membership lists built from historical index constituents. Delisted names are retained with terminal values. Walk-forward uses historical constituents only, never today's index projected backwards.
- **Point-in-time store (Stage 12).** A read as of date t provably returns only information knowable before t. Fundamentals and announcements are keyed by publication time, not event time.

5.6 The Indian transaction cost model

The cost model is built at Stage 6, before any feature, because the largest risk to the project is not that the architecture fails but that a sophisticated system is built and then found to have no edge after Indian frictions.

Component	Delivery (swing, long-term)	Intraday
STT	0.1% on both sides	0.025% on the sell side only
Brokerage	Discount-broker schedule, assumption documented	same
Exchange charges	NSE published schedule	same
GST	On brokerage plus transaction charges. A tax on the fees, easy to omit, and omitting it flatters results.	same
Stamp duty	Buy side, state-varying (Maharashtra rates used)	same
SEBI turnover fee	Small but present	same
Impact cost	NSE published, per security	same
Round-trip total	**About 0.25 to 0.30%**	**About 0.06 to 0.10%**

Intraday is *cheaper* per round trip than delivery in India, because STT is 0.025% on the sell side only. What makes intraday hard is signal-to-noise and execution timing, not cost.

All rates live in a **date-versioned table**, so a backtest over 2011 applies 2011's STT. STT did not exist before 1 October 2004. The rates for stamp duty and exchange charges before that date are carried backwards and are flagged in the code as not independently verified.

6. Features, override gates and the deterministic scorer

6.1 Feature blocks

Block	Content	Notes
Price and technical	Returns, ATR, moving averages, RSI and similar indicators	Computed by a dependency-free, database-free function so it can be unit-tested without a live database
India block	Delivery %, F&O open interest and rollover, FII/DII flows, promoter pledging, circuit and ban status	Delivery masked, never zero-filled, where unavailable
Macro and regime inputs	India VIX, Nifty 500 breadth, USDINR change, repo rate, CPI, FII flows	Feed the regime model
Knowledge-graph features	Event type, polarity, magnitude, horizon and confidence from extraction; propagated supplier and group effects	Extractive only
Deterministic scorer	Four-layer weighted score, z-scored per date	Feature and fallback, not a decision-maker

6.2 Override gates

News gate, volatility gate, drawdown gate and event-proximity gate run before scoring. They are cheap, auditable, and catch failure modes that no learned model reliably catches. India-specific hard gates are retained: circuit limits, F&O ban periods, the liquidity cliff below the top 200, corporate-action frequency and promoter pledging.

7. Language and knowledge layer

7.1 Why a structured extractor instead of FinBERT

The case against FinBERT is not language: the corpus is almost entirely English. It is that FinBERT was trained on US text, so concepts such as promoter pledging, upper circuit, F&O ban, PSU/NBFC classification and QIP appear to it as unfamiliar tokens. A single sentiment number in [-1, +1] also discards the event type, the entity, the direction, the magnitude and the horizon, all of which the knowledge graph needs. Recent work finds that small instruction-tuned LLMs can match or beat FinBERT with little or no task-specific data.

7.2 The extractor

An instruction-tuned open model, **Qwen3-4B** with LoRA/QLoRA fine-tuning (confirmed to fit the 4 GB local card for inference), emits one structured record per article or announcement.

Field	Content
`entity`	Resolved NSE symbol
`event_type`	One of about 18: results_beat, results_miss, guidance_up, guidance_down, order_win, capex_announcement, rating_upgrade, rating_downgrade, regulatory_action, promoter_buy, promoter_sell, pledge_increase, pledge_release, mna, litigation, management_change, qip_fundraise, other
`polarity`	-1, 0 or +1
`magnitude`	minor, material or transformative
`horizon`	immediate, short, medium or long
`confidence`	0 to 1
`source_span`	Verbatim text that supports the extraction
`related`	Co-mentioned entities, which populate graph edges

> **THE RULE THAT KEEPS THE AGENT LAYER HONEST: EXTRACTIVE, NOT GENERATIVE**
>
> Facts written to the knowledge graph must be extractive and verifiable. “Company A supplies Company B”, extracted from a related-party disclosure with a `source_span`, can be checked against the original. “Sector sentiment is deteriorating” is an LLM opinion with no ground truth, and it never enters the feature matrix. Extraction can be validated by hand-labelling 500 articles and reporting precision and recall per event type. Judgment cannot be validated.

7.3 Knowledge graph

Neo4j stores entities, events and relations. Agents write to the graph and never to the signals table. The graph feeds the model in two ways: relational features, and two of the six edge types in the stock graph (supply chain and co-mention). Every graph snapshot used in training is built from trailing data only.

7.4 Hardware routing

Stage	Where	Note
SSL pretraining	Cloud GPU (Kaggle T4 16 GB or Colab)	The long runs. Checkpoint every epoch, because free sessions time out.
LLM LoRA fine-tune	Cloud GPU	QLoRA
LLM batch inference	Cloud or local	Nightly over new items. The historical backfill is a one-time bulk job.
Supervised fine-tune	Local 4 GB	Small task head (about 500k parameters)
Nightly inference	Local 4 GB	700 stocks, one forward pass

NLP inference and model training are never co-resident on the 4 GB card. Extraction is a nightly stage that finishes before any training begins.

8. PRISM: the representation

PRISM (Pretrained Relational Index-neutral Signal Model) is everything that turns data into a score-ready representation. It is kept from the v4 design except where Section 9 says otherwise.

8.1 Cross-market self-supervised pretraining

The encoder is trained before it sees any ranking label, on a task that needs no labels at all.

Item	Specification
Objective	Masked patch reconstruction. A price sequence is cut into patches of 5 trading days, a random 40% are masked, and the encoder reconstructs them from context. Loss: mean squared error over masked positions only.
Why it works	Reconstruction forces the encoder to learn volatility clustering, mean reversion and volume-price coupling, which are the properties that make masked patches predictable.
Corpus	India NSE bhavcopy (about 1,500 tickers, including names outside the final universe, because illiquid names are still valid sequences), the SignalForge US dataset (about 800 tickers), and about 40 index series
Windowing	Overlapping 250-day sequences with stride 5: on the order of one million sequences, three orders of magnitude more than the 5,000 effective supervised samples
Encoder	Patch transformer, 6 layers, d_model 256, 8 heads, about 5 million parameters
Why SignalForge helps	Unlabelled price sequences carry no market-specific label meaning. What transfers is the shape of financial time series. The first project's data becomes the second project's pretraining corpus.

> **PRETRAINING LEAKAGE: THE FAILURE UNIQUE TO THIS DESIGN**
>
> The pretraining corpus must be split **by time, not by ticker**. An encoder pretrained on sequences running to 2026 and then fine-tuned on a fold that tests 2015 has seen the future of that fold, and every result would be silently invalid. For a fold testing year Y, both pretraining and fine-tuning use data ending before Y minus the embargo. In practice this means one encoder per fold. Cutting this corner is the most damaging shortcut available in the architecture.

8.1.1 WFP: warm-started walk-forward pretraining

The problem. Training every fold's encoder from scratch costs 12 to 48 hours on a T4. Over up to 25 price folds that is several hundred GPU-hours, several times the free Kaggle allowance of about 30 GPU-hours a week. As specified in v5, the plan did not fit its own compute budget.

The change. The first fold is pretrained from scratch. Each later fold k starts from fold $k-1$'s checkpoint and continues pretraining on the sequences added since, up to fold k's cutoff (test start minus embargo), with a short replay sample of older sequences so earlier eras are not

forgotten. Fold $k-1$'s checkpoint only ever saw data ending before fold $k-1$'s cutoff, which is earlier than fold k's, so the chain never sees the future of any fold.

Item	Specification
First fold	From scratch, full schedule
Later folds	Continue from the previous fold's checkpoint. New data since the last cutoff plus a 20% replay of older sequences. Fixed step budget, set at Stage 21 from the loss curve.
Leakage rule	Every ancestor checkpoint's data cutoff is at or before the current fold's cutoff. Enforced by invariant N14 from the checkpoint lineage recorded in MLflow.
Expected saving	Later folds should need a small fraction of the from-scratch time. The real figure is measured at Stage 21, not assumed.
Check that it is not hurting	Ablation A18: warm-started against from-scratch on three folds, compared on held-out reconstruction loss and on downstream rank IC
Fallback	If A18 shows a material loss, train from scratch only on the folds used for the headline PRISM results and report the reduced fold count

8.2 Heterogeneous temporal graph (R-GAT)

Stocks are nodes. Six edge types are rebuilt monthly from trailing data only.

Edge	Source	Contribution
same_sector	NIC / GICS classification	Baseline economic linkage
same_promoter_group	NSE/BSE promoter disclosures: about 35 groups and 150 entities	The India-specific edge. Tata, Reliance, Adani, Birla, Bajaj, JSW and Vedanta members reprice together on group news. In January 2023 one report moved ten Adani-group tickers. There is no US analogue.
supply_chain	Knowledge graph, from related-party disclosures and news	Propagates a supplier's results to its customers. This edge is what makes the agent layer contribute measurably rather than decoratively.
corr_knn	Top 8 by 120-day return correlation, excluding same-sector	Co-movement the taxonomy misses
index_comembership	Nifty-family membership	Pure flow effect from index-fund rebalancing
co_mention	Knowledge graph, from GDELT co-occurrence in one article	Emergent, event-driven association

The model is a **relational graph attention network with two layers**. Two only: with 700 nodes and dense sector edges, two hops already reach most of the graph, and a third layer smooths node representations toward one global average, which rediscovers the market factor that the label design removes. Attention is used instead of fixed weights because the right weight on a promoter-group edge depends on state: group members track each other tightly

during group events and loosely otherwise. A promoter-group dummy variable in a tree model encodes membership. Attention encodes *propagation*, whose strength varies.

8.3 Regime model

A 3-state Gaussian hidden Markov model over India VIX, FII net flow, Nifty 500 breadth and USDINR change describes the market regime. It is fitted on training data only.

> **THE BUG THAT INVALIDATES REGIME RESULTS: SMOOTHED STATES**
>
> At inference the model must use the **forward algorithm**, which gives the *filtered* state probability using observations up to time *t*. Viterbi decoding and smoothed posteriors condition on the whole sequence, including the future. A backtest built on them looks spectacular and does not survive a week live. The common library trap: `hmmlearn`'s `predict_proba` runs forward-backward and returns **smoothed** posteriors. It is forbidden here, and the forward recursion is implemented explicitly.

In v5 the regime model no longer routes to hard-selected heads. Its filtered posterior is used as a set of soft weights in VN-ListNet (Section 9.2) and as part of the RW-CS calibration weights (Section 9.3).

8.4 Why a small supervised head on a large encoder

The pretrained encoder may be large because it learns from a million unlabelled sequences. The task head that maps it to a score sees about 5,000 effective samples and stays small. This split is the most defensible design decision in the project, and it is what makes "we trained for three days" a real claim rather than a boast.

9. VANTA: the decision pipeline

VANTA replaces PRISM's labels, ranking loss and publish gate. The encoders, graph, data layer and language layer stay. This section is the authoritative specification.

9.0 The pipeline and its components

VANTA is a pipeline: labels, then ranking, then selection, then trade levels, then tiers. It contains four named components (Section 1.1). Each one has its own specification, its own invariant tests, and its own ablation, so it can be kept or dropped on evidence.

Part	Component	Section	Kept only if
A	**NEL**, net executable labels	9.1	Net return per published signal beats gross labels (A2)
B	**VN-ListNet**, volatility-normalised ranking with soft regime gating	9.2	It beats plain ListNet (A3) and is not beaten by B1 through the same gate; otherwise B1 becomes the ranker and the result is reported
C	**RW-CS**, regime-weighted conformal selection	9.3	Its realised FDR by regime is closer to q than age-only and uniform weights (A16)
C	**FCC**, full-candidate calibration	9.3	Always kept. It is a correctness rule (N8). A17 measures how much bias published-only calibration would have introduced.
D	ATR trade levels	9.4	A11 shows stops and targets do not destroy expected value

Every component's settings are chosen on validation folds only and recorded in the version tags (Appendix A.4).

9.1 Part A: NEL, net executable labels

For every (stock i, decision date t), the system simulates the exact trade that would have been published at t.

Rule	Specification
Decision time	After the close of t. The weekly decision date is the last session of the week.
Entry	Raw open of session $t+1$. **No fill** if the stock opens locked at its upper circuit, defined as: the symbol is on that day's price-band hit list as an upper hit **and** raw open equals raw high.
Stop	Entry minus k_s times ATR(14) at t. If price gaps through the stop at the open, the fill is at the open.

Rule	Specification
Target	Entry plus k_t times ATR(14) at *t*. If price gaps through the target, the fill is at the target (conservative: no windfall).
Same-day ambiguity	If stop and target both sit inside one day's range, assume the **stop hit first**. Daily bars cannot resolve the order, so the conservative choice is used.
Lower-circuit lock	A session is lower-locked at close if the symbol is on the hit list as a lower hit **and** raw close equals raw low. A stop triggered inside a lock exits at the close of the first session that is not lower-locked, and the delay is recorded.
Time exit	Close of session $t+h$
Corporate actions	Returns use Stage 4 adjusted prices. Lock detection always uses raw prices against the band.
Delisting	Exit at the last traded price (the Stage 5 terminal value)
Cost	Stock-specific round trip from the Stage 6 model at trade-date rates, plus impact cost. Entry and exit legs are stored separately.
Label	y = payoff minus round-trip cost, a net return

k_s and k_t are chosen per horizon on validation folds only, then frozen per fold and versioned.

Design choices worth knowing.

- **Incumbency is not baked into labels.** An early proposal let names already published last week skip the entry cost inside the label. That makes the training target depend on the model's own past publications, which do not exist historically and change whenever the model changes, so the target would become circular. Labels stay policy-free: every row is a full round trip, and `cost_entry` is stored separately so an incumbent's net return can be computed later. The accounting layer charges the entry leg once per position (invariant N12).
- **No-fill handling is a training configuration, not a stored value.** Default: a penalty value. It stays configurable so it can be ablated.
- **Lock detection uses only the hit lists.** Inferred price bands were measured at 87.8% accuracy, which is not good enough, so rows with no hit-list coverage are excluded from primary results.
- **Scope.** Horizons of 5, 20 and 60 sessions on daily bars.

9.2 Part B: VN-ListNet, volatility-normalised listwise ranking

1. **Warm start.** A ridge regression or a small MLP on the net labels, on top of encoder outputs (or on the raw feature matrix for baselines).
2. **Fine-tune.** ListNet (Plackett-Luce) loss over each decision date's candidate list. The gain is y divided by (ATR% times the square root of h), winsorised at the 2nd and 98th percentiles of the **training fold only**.
3. **Why normalise.** Top-heavy listwise losses on noisy returns favour high-volatility "lottery" stocks. In the synthetic runs, removing the normalisation dropped simulated Sharpe from 2.42 to 0.23. Ablation A3 tests whether the same effect appears on NSE data.

4. **Soft regime gating.** Score = sum over regimes *k* of the filtered posterior π_t(k) times head_k(z). The weights are posterior probabilities, not learned parameters, so the rule against learned gating networks still holds in spirit.

9.3 Part C: RW-CS and FCC, the publish gate

The gate follows conformal selection (Jin and Candès 2023a) with non-exchangeable weighting in the spirit of Barber et al. (2023).

Calibration set at decision date T. Out-of-sample (score, realised net return) pairs for **all scored candidates** from the last 52 weeks, not only those that were published, because restricting to published names introduces selection bias (invariant N8). A row enters only after its horizon has passed and its outcome is known (invariant N5). For the 60-session horizon the effective window is shorter, because recent rows have not matured.

Weights.

w_i = exp(-age_i / τ) times exp(-‖π_i - π_T‖₁)

Age is in weeks, π is the 3-state filtered posterior, and the distance is the L1 norm. The test point carries weight 1. A candidate gets more weight in the calibration if it is recent and if its market regime resembled today's. The decay scale τ is chosen per horizon on validation folds from {13, 26, 52} weeks (default 26), then frozen and recorded in `gate_version`. It is never chosen by looking at test folds.

Computation. Sorting the calibration scores once per decision date and keeping a running sum of the weights of losing rows gives every candidate's p-value in O((n + m) log n), for n calibration rows and m candidates. For about 175,000 rows and 700 candidates this takes well under a second on a CPU, so the gate adds nothing to the GPU budget.

p-value for each test candidate *j* (null hypothesis: the trade loses after cost, y ≤ 0):

p_j = (1 + sum over i of w_i · 1[y_i ≤ 0 and s_i ≥ s_j]) / (1 + sum of w_i)

This is the clipped-score conformal p-value of Jin and Candès with conservative tie handling and weights.

Selection.

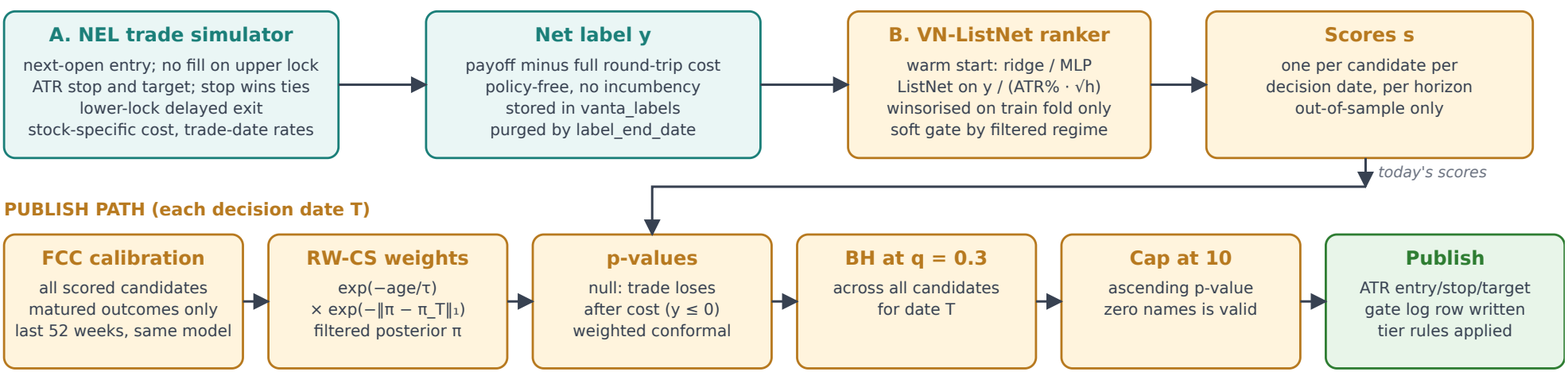
1. Apply Benjamini-Hochberg at q = 0.3 across all candidates for date T.
2. Keep at most 10 names, in ascending p-value order, ties broken by score.
3. Zero published names is a valid outcome.

What the gate promises, stated precisely.

- **Target.** On average, at most q of published signals lose after cost.
- **Guarantee status.** Exact false-discovery-rate control needs exchangeability. Three things break it here: financial time series are not exchangeable; the weights depend on observed data (π_T), which takes the method outside the fixed-weight bound of Barber et al.; and plain BH on weighted conformal p-values is not covered by Jin and Candès's result (their weighted-selection paper adds a pruning step for this reason). The cap at 10 also keeps a subset of the BH set, which the BH guarantee does not cover.
- **Therefore** the promise is an engineering target. It is validated empirically and reported by regime. It is not a theorem, and the project never describes it as one.

- **Measurement.** FDR is the mean of the weekly false-discovery proportion, with weeks that publish nothing counted as 0. It is reported next to the pooled loss share (total losers divided by total published). The two can differ a lot when weekly publish counts vary.
- **Fallback.** The Kish effective sample size, $(\Sigma w)^2 / \Sigma w^2$, is logged for every decision. If regime weighting shrinks it below a threshold (proposed: 200), the gate falls back to age-only weights and logs that it did.

VANTA: from simulated trades to a published list



What the gate promises, stated precisely

Target: on average, at most q = 30% of published signals lose after cost.

Exact FDR control holds only under exchangeability. Markets are not exchangeable, the weights depend on observed data, and plain BH on weighted conformal p-values is outside the published theorems. The cap at 10 also keeps a subset of the BH set. So q is an engineering target, measured as realised FDR by regime and reported with the pooled loss share. It is not a theorem.

VANTA from simulated trades to a published list

9.4 Part D: trade levels

Labels simulate stops and targets at k_s and k_t times ATR. If the published levels came from a different source, the labels would not describe the published trade. So the published levels are the same ATR multiples: entry at next open, stop at entry minus k_s·ATR, target at entry plus k_t·ATR. The v4 quantile heads leave the decision path and may be kept only as a displayed “expected range” under ablation A11. If they are kept, the quantile-crossing check (a lower quantile never above a higher one) still applies to them.

9.5 Evidence so far, and how much to trust it

Configuration (synthetic data, 6 seeds)	Result
Baseline: v4-style ranker and per-stock gate	Sharpe 1.05, net 0.36% per signal, 47% of signals lost
A and B only (net labels and normalised ranking)	Sharpe 2.42
A and B without volatility normalisation	Sharpe 0.23
Full VANTA (A, B and gate)	Win rate 60%, net 1.10% per signal, 14% of signals lost (pooled share)

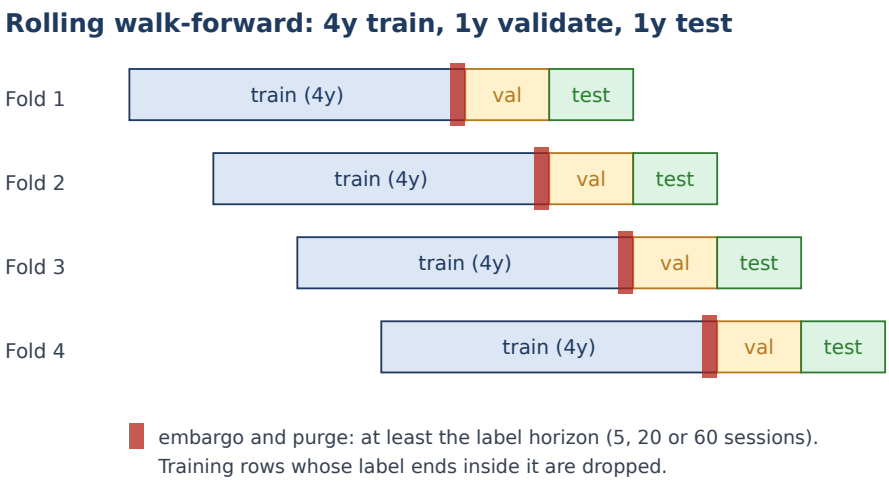
> **READ THESE NUMBERS WITH CARE**
>
> The synthetic data-generating process may itself contain the effects that VANTA is designed to fix, such as heavy idiosyncratic volatility tails. These figures are design evidence only. They do not transfer automatically to NSE data. Milestone M1 and ablations A2, A3 and A10 decide.

10. Training and validation protocol

10.1 Four training stages per fold

#	Stage	Data	Duration	Output
1	SSL pretraining (WFP)	About one million unlabelled sequences, ending before the fold boundary	First fold 12 to 48 hours; later folds warm-started from the previous fold (Section 8.1.1), measured at Stage 21	Encoder checkpoint
2	Supervised fine-tune	Fold training window, net executable labels. Ridge or MLP warm start, then ListNet.	1 to 3 hours per seed, 10 seeds	Ranker
3	Regime model and soft gating	HMM fitted on the training window. Filtered posterior computed by forward recursion.	Under 1 hour	HMM and gating weights
4	Calibration store	Out-of-fold scores for the preceding 52 weeks, written before the gate runs, with outcomes added as they mature	Minutes	`vanta_calibration` rows

10.2 Walk-forward discipline



Rolling walk-forward with embargo

The window is **rolling, not expanding**: four years to train, one to validate, one to test, rolled by one year. Rolling matters because Indian market structure has broken repeatedly. Under an

expanding window the 2025 model is partly fitted on 2006 microstructure, which is contamination, not data.

- **Embargo and purging.** The embargo is at least the label horizon, per horizon (5, 20 or 60 sessions), and is a configuration parameter, never a constant. Training rows whose label ends inside the embargo (`label_end_date` at or after test start minus embargo) are dropped. In multi-horizon training the longest horizon governs, which is a real cost of multi-task learning and is stated rather than ignored. (A 2-day embargo against a 3-day label was an earlier mistake: it leaks.)
- **Seeds.** Ten, not five. Seed variance at this sample size is large. All figures are mean and standard deviation across seeds and folds. A single-seed number is noise.
- **Nested search.** Hyperparameters are tuned inside the training window so the test fold is never tuned on.
- **Statistics fitted on the training fold only.** Normalisation, winsorisation, k_s and k_t, the HMM and gain scaling all use training-fold data only (invariants N3 and N10).

10.3 Where compute helps and where it does not

Spend	Helps?	Why
SSL pretraining, once per fold	Yes	Learns from a million unlabelled sequences. The dominant compute line, and the one that earns its cost.
Warm-starting each fold from the previous one (WFP)	Yes	Keeps per-fold pretraining leak-free while removing most of its repeated cost. Checked by A18.
10-seed ensembling	Yes	Turns a noisy point estimate into a distribution. Embarrassingly parallel.
Nested hyperparameter search	Yes	Prevents tuning on the test fold, the quiet way most reported results inflate
Historical LLM extraction backfill	Yes	One-time bulk job that unlocks news features across the whole walk-forward
Larger supervised head	No	5,000 effective samples: more parameters fit noise
More epochs on the task head	No	Early stopping on validation rank IC governs
Neural architecture search	Actively harmful	A model free to redesign itself against 5,000 samples finds noise and reports it as structure

11. Evaluation

For a project of this kind the evaluation chapter carries more weight than the architecture chapter. A modest model measured honestly is a stronger submission than a sophisticated model measured carelessly.

11.1 Metrics

Metric	Definition	Purpose
Rank IC	Daily Spearman correlation of predicted and realised cross-sectional rank	Primary measure of ordering skill
IC-IR	Mean rank IC divided by its standard deviation	Consistency: IC 0.03 every day beats IC 0.06 half the time
NDCG@10	Top-weighted ranking quality at 10	The product metric, because the output is a shortlist
Net return per published signal	Mean net return of published signals after full costs, with stops and targets active	The number that decides whether the product is useful
Realised FDR	Mean weekly false-discovery proportion, zero-publish weeks counted as 0, by fold and regime	Evidence for the gate's promise
Pooled loss share	Total losing signals divided by total published	Reported next to FDR
Calibration coverage	Realised loss rate among calibration-eligible candidates against the gate's implied rate	Evidence for the calibration claim
Publish-count distribution	Weekly counts, share of zero-publish weeks, how often the cap binds	A healthy system abstains sometimes
Hit rate, realised R, stop-out rate	Path-dependent outcomes with stops active	Rank IC alone no longer suffices once stops exist
Extraction precision and recall	Per event type, against 500 hand-labelled articles	Evidence that the agent layer is trustworthy

11.2 Baselines

- **A0, momentum 12-1.** The dumb baseline. Many published ML equity results fail to beat it net of costs.
- **A1, SignalForge port.** The predecessor: deterministic score with an XGBoost binary classifier, thresholded. Does the reformulation beat it on identical Indian data under identical conditions?
- **B1, CatBoost on the same features and splits, trained on net labels and passed through the same VANTA gate.** This is the main baseline. Published gains from stock graph networks and from pretraining on financial series are small and inconsistent, and finance pretraining at best ties a well-tuned gradient-boosted model. Comparing PRISM plus VANTA against B1 isolates whether the encoders earn their cost.

11.3 Ablation suite

Each ablation isolates one claim. This is what makes the project research rather than a demonstration.

ID	Comparison	Question
A0	Momentum 12-1	The baseline everything must beat
A1	SignalForge port	Does the reformulation beat the predecessor?
B1	CatBoost, same features and splits	**Main baseline**
A2	Gross against **net** labels	Do executable net labels improve net return per published signal?
A3	With against without **volatility normalisation**	Does the lottery-stock effect seen in synthetic runs appear on NSE?
A4	Remove the US corpus	Does cross-market transfer help, or only pretraining?
A5	Remove multi-horizon training	Does multi-task supervision improve label efficiency?
A6	Remove the graph	Does relational structure contribute?
A7	Remove promoter-group edges only	The India-specific claim. Degradation here shows a market-structure effect with no US analogue.
A8	Remove knowledge-graph features	Does the agent layer contribute measurably, or only decoratively?
A9	Hard-routed heads against soft gating against no regime	Does soft regime gating help?
A10	Gate arms: (a) ungated top 10, (b) per-stock conformal gate (v4), (c) VANTA gate	Does the new gate beat both?
A11	ATR-multiple levels against hold to horizon, and quantile levels as display only	Do stops and targets add or destroy expected value, per horizon?
A12	Full system	The system as specified
A13	LightGBM lambdarank	Isolates the ranking objective
A14	Remove SSL pretraining, tested against B1	Does pretraining earn its cost?
A15	q in {0.2, 0.3, 0.4}	Sensitivity of the promise
A16	Gate weights: uniform, age only, age times regime, age times regime with weighted-selection pruning	Is RW-CS worth it? Does pruning change realised FDR?
A17	Calibration on all scored candidates (FCC) against published names only	How much optimistic bias would published-only calibration introduce? FCC stays in either case.
A18	WFP warm-started pretraining against from-scratch, on three folds	Does warm-starting cost accuracy, and how much time does it save?

A1, A3, A7 and A14 are the ones that matter most, with A2 and A16 as the direct tests of the two primary contributions (NEL and RW-CS). A negative result on any of them is still a result, and a

well-measured negative finding outranks an unmeasured positive claim. Every gated arm is reported by fold and by regime (regime is bucketed by the argmax of the filtered posterior).

11.4 Era analysis and expected results, stated in advance

Rank IC is reported by era across the whole walk-forward, covering the stress windows of 2000, 2008, 2013, 2016 (demonetisation), 2018 (IL&FS and the NBFC crisis), 2020 (COVID) and 2021 (retail boom). Reporting through demonetisation and the NBFC crisis is a harder test than any US equivalent. If early-period IC differs from recent-period IC, the project has empirically shown microstructure non-stationarity in Indian equities, which turns a data limitation into a finding.

> **NUMBERS TO EXPECT, STATED BEFORE TRAINING**
>
> A rank IC of 0.03 to 0.05 for swing horizons is genuinely good. Longer horizons may run slightly higher. **An IC above 0.10 is a bug**, and the response is to open the invariant tests, not to celebrate. A model that reports 85% directional accuracy has leakage, which is a near-certainty to diagnose, not a possibility to weigh. Stating this in advance is a form of pre-registration: it makes the eventual result more credible whatever it is.

11.5 Milestone M1: the first real-data test of the new layer

Purpose. Test the label simulator (Part A) and the gate (Part C) on real NSE data before any new model is trained. The scores are 12-1 momentum, recomputed weekly, which is already validated (the Stage 7 GO used monthly rebalancing). Any failure at M1 therefore belongs to A or C, not to a model.

Setup. Horizons 5 and 20. k_s in {1.0, 1.5, 2.0} and k_t in {1.5, 2.0, 3.0} are chosen on the first three years of the window only, then frozen. The window starts at the first price-band hit list on disk. Two gate arms: age-only weights (needs no HMM) and age times regime weights (needs the filtered HMM and covers only the later part of the window). Comparisons: ungated weekly top 10 and the v4 per-stock gate.

Pre-registered GO criteria, fixed before running.

#	Criterion	Threshold
1	All invariants N1 to N13 pass on real data	Required
2	Realised FDR over the evaluation window	At most q + 0.05 (0.35)
3	No calendar year has realised FDR above	0.5
4	The gate is not degenerate	Publishes in at least 20% of weeks
5	Net return per published signal, gated arm	At least the ungated top 10, with a weekly block-bootstrap 95% confidence interval

Reading the result. If momentum has little weekly edge, a gate that mostly abstains **passes** criteria 2 and 3: the gate is doing its job. If criteria 2 or 3 fail, the gate is fixed before any new model is trained. The Stage 7 result (about 4.6% net annualised long-short decile spread, monthly) measures something different and is not evidence for or against M1.

12. Invariants and the silent-failure catalogue

12.1 Rules that are never broken

- The LLM never emits a decision. PRISM and VANTA decide; the LLM narrates.
- No trade execution. Signalling and information only.
- Every scheduled job is wrapped so that one failure never stops the scheduler.
- All secrets live in `.env` and are read through `config.py`. One database path and one model path, fixed there on day one.
- Walk-forward uses historical constituents only.
- Every signal stores its full feature snapshot as JSON, and every signal shown to a user carries a disclaimer.
- Agents write to the knowledge graph and never to the signals table. Graph facts are extractive and carry a `source_span`.
- Pretraining respects the fold boundary, including every warm-start ancestor checkpoint. Embargo is at least the label horizon, per horizon.
- NLP inference and training are never co-resident on the 4 GB card.
- No smoothed or Viterbi regime states anywhere. No learned gating network. No architecture search.
- No unmatured outcomes in the calibration store, and no mixing of model versions inside one calibration set.
- No incumbency logic inside training labels.
- The gate’s q is never described as a guarantee.
- No zero-filling of missing features.

12.2 Silent failures: bugs that look like success

These raise no error and often make results *better*. The only defence is a test that asserts the invariant directly, on a hand-built input with a known answer, and that is proved to fail against a deliberately broken implementation before it is trusted.

#	Area	Failure	Symptom	Test
1	Corporate actions	Unadjusted splits and bonuses	Spurious -50% returns; the ranker learns bonus calendars	No adjusted return outside its price band without an action record
2	Graph	Graph snapshot includes future edges	Nothing errors; IC quietly rises	Every snapshot built from trailing windows only. A synthetic future-dated edge must not change a prediction.
3	Normalisation	Full-history statistics	Invisible; IC roughly doubles	Cross-sectional z-score per date; assert no full-history statistics

#	Area	Failure	Symptom	Test
4	Regime	Viterbi or smoothed states	Backtest spectacular, live results noise	Forward recursion only (N7)
5	Pretraining	Encoder saw the test fold, directly or through a warm-start ancestor	Every fold silently invalid	Encoder checkpoint cutoff, and every ancestor's cutoff, earlier than fold test start minus embargo (N14)
6	Labels	Horizon off by one	The 5-day head trains on 4-day returns; loss looks normal	Hand-labelled toy path: each horizon's label equals the manually computed forward return
7	Levels	Quantile crossing (only if display quantiles are kept)	A stop above entry	Lower quantile at most the higher quantile on every displayed signal
8	Calibration	Window overlaps the test fold or holds unmatured outcomes	Coverage looks right in backtest, degrades live	N5, N8 and N9 below
9	Costs	Wrong-side or stale cost	Signals publish that should abstain	Cost function matches position side and changes across STT regime dates (N6)
10	Explanation	LLM states a fact not in the evidence	Fluent, plausible, unsupported	Every named driver in the text appears in the SHAP list or retrieved evidence; otherwise fall back to a template
11	Validation	Date-type mismatch in queries	Empty windows pass silently	Compare ISO strings, never integers; assert row counts per window
12	Verification	A check queries the wrong table	A stage passes without testing its own output	Every stage check queries the table that stage writes

12.3 VANTA and PRISM invariant tests N1 to N14

ID	Invariant	Test
N1	No fill on a locked open	A day on the hit list as upper with open equal to high gives `NO_FILL_UPPER_LOCK`, no exit price and no payoff

ID	Invariant	Test
N2	Stops respect lower-circuit locks	Stop triggers on day d. Days d and d+1 are lower-locked at close, d+2 is not. Exit is at d+2's close with `STOP_DELAYED_LOWER_LOCK` and a delay of 2.
N3	Same-day ambiguity resolves to the stop	Both levels inside one day's range gives exit at the stop
N4	Gap fills	A gap below the stop fills at the open. A gap above the target fills at the target.
N5	The calibration set never holds an unmatured outcome	Every calibration row has `matured_at` at or before T. An injected row with `matured_at` of T+1 is caught. Adding outcomes after T never changes p-values at T.
N6	Labels use the cost model of the trade date	Output changes across a known STT rate-change date
N7	Regime weights use filtered posteriors	The posterior at *t* computed on data up to *t* equals the posterior computed on the full series. Appending the future changes nothing.
N8	Calibration covers all scored candidates	Calibration row count for a week equals candidates scored, not names published
N9	Calibration and test share model version, candidate filter and out-of-sample status	A mismatched row is rejected
N10	Normalisation and winsorisation come from the training fold only	Changing test-fold labels does not change training gains
N11	Purging	No training label with `label_end_date` at or after test start minus embargo
N12	Cost is charged once for incumbents	A name selected in weeks w and w+1 shows one entry leg and one exit leg in the ledger
N13	p-value mechanics	p is in (0, 1] and non-increasing in score. Equal weights reduce to the unweighted conformal p-value. Published names are a subset of the BH set, at most 10, with `cap_binding` logged.
N14	WFP checkpoint lineage	Every ancestor of a fold's encoder checkpoint has a data cutoff at or before that fold's test start minus embargo. A checkpoint whose lineage contains a later cutoff is rejected.

Delivery (Layer 8) and self-improvement (Layer 9) are excluded from this style of test on purpose. They are operational rather than statistical and are verified by integration testing: ten unattended runs, tier checks and log audits.

13. Platform and delivery

13.1 Technology stack

Area	Choice	Rationale
Data	NSE bhavcopy (scraped), GDELT 2.0 GKG via BigQuery, NSE/BSE announcements, NSDL, RBI DBIE, MOSPI	Section 5. Kite Connect, Dhan or SmartAPI only for the deferred intraday work.
Models	PyTorch, PyTorch Geometric, CatBoost (main baseline), LightGBM (ablation), hmmlearn (parameter fitting only; inference is an explicit forward recursion), SHAP, Optuna	Encoder, graph network, baselines, attribution, nested search
Gate	Custom weighted conformal selection with Benjamini-Hochberg	No library implements this exact combination, and the p-value mechanics are covered by tests N5 and N8 to N13
Language	Qwen3-4B with LoRA/QLoRA, LangChain, LangGraph	Extractive extraction. A fixed, inspectable agent loop.
Memory	Neo4j, ChromaDB or pgvector, Groq API	Knowledge graph, vector store, and explanation generation. Explanation is off the decision path, so model choice carries no decision risk.
Platform	Prefect, MLflow, PostgreSQL, Parquet, Redis, Docker Compose	One nightly DAG and one morning DAG. MLflow is the registry and audit trail: the ablation suite is unreproducible without it.
Compute	Kaggle T4 or Colab (pretraining, LLM fine-tune); local RTX 3050 Ti 4 GB (supervised fine-tune, inference)	Checkpoint aggressively; free tiers have session limits
Delivery	FastAPI, Next.js with TradingView Lightweight Charts, React Native with Expo, WebAuthn and Argon2id, Razorpay	Section 13.2

> **DELIBERATELY ABSENT**
>
> No Kafka, no Kubernetes, no vector-database SaaS, no feature-store SaaS. Serving is one cached snapshot, identical for every user, and the pipeline is two scheduled batches. An architecture sized to its problem defends better than one sized to a job description.

13.2 API, web and mobile

- **One API, two clients.** FastAPI serves Lite, Pro and Max endpoint families. The Next.js web app and the React Native (Expo) mobile app are two clients of the same API, not two systems. Neither client holds decision logic.
- **Tiers are enforced on the server.** A Lite token calling a Pro endpoint gets a 403 with no body.

- **Charts.** TradingView Lightweight Charts (Apache-2.0) in the web client.
- **Authentication.** Passkeys (WebAuthn) with Argon2id for any stored secret. On mobile, Face ID or the Android biometric prompt unlocks a device-held key locally. No biometric data is ever sent to or stored on the server. Payments use Razorpay with UPI Autopay for recurring INR, in test mode until the registration question is resolved.
- **Immutable signal log.** Every published signal is written once and never edited. The public paper-trading record is generated from this log.

13.3 Mobile delivery

Why Expo. Expo's managed workflow handles code signing, store submission tooling (EAS Build and EAS Submit) and over-the-air updates without hand-built Xcode and Android Studio pipelines. For a small team shipping to two stores under a deadline, this reduces the surface where an AI coding agent can get something subtly wrong. Bare React Native remains an escape hatch.

Store	Requirement	LYNX-specific consideration
Apple App Store	Apple Developer Program enrolment. Review covers functionality, data use and financial-app scrutiny.	LYNX names entry, target and stop-loss. The finance-app guidelines require a clear statement that content is informational, not investment advice. It must be visible in the store listing itself, not only inside the app.
Google Play	Play Console account, Data Safety form, financial-services content declaration	The declaration must state that the app does not execute trades and does not give personalised investment advice. It is a distinct submission step, separate from the in-app disclaimer.
Both	A privacy policy URL	It must state what is collected (authentication credentials and subscription status) and confirm there is no trade-execution capability.

> **THE SEBI RESEARCH ANALYST QUESTION**
>
> Payment in exchange for entry, target and stop-loss levels is the textbook form of a recommendation, and SEBI Research Analyst registration may apply once money is involved. Both stores require an explicit financial-content declaration at submission, so the question must be **resolved**, not just documented, before either submission, not only before public launch. This is an open risk (Section 16).

Offline behaviour. LYNX publishes at most ten signals a day, generated once nightly. The mobile client caches the last published set and shows it with a visible timestamp instead of a blank screen. There is no offline computation: the model never runs on a device, so the rule that the model is the only decision-maker holds on every surface.

Updates. JavaScript-layer changes (copy, UI fixes) ship over the air through Expo. Native-layer changes (new permissions or modules) still go through full store review.

14. Environment and setup guide

This section is the working checklist for bringing up a development machine. Paths and commands follow the repository as it exists; items marked *later stage* are not available yet.

14.1 Prerequisites

Item	Needed for
Python 3.12	All backend code. CI also uses 3.12.
Docker Desktop with Compose	Postgres, Redis and Neo4j
Git	Source control and CI
CUDA-capable GPU (optional locally)	Stage 22 supervised fine-tune and nightly inference. Pretraining uses a cloud GPU.
Kaggle and Google accounts	Cloud GPU runs, and BigQuery for GDELT (later stage)
A coding assistant (Claude Code in the IDE)	Development. It loads the project memory file automatically.

14.2 Repository layout (as it exists so far)

```
config.py                canonical config: DATABASE_URL, EMBARGO_DAYS, paths, get_db_engine
docker-compose.yml       Postgres 16, Redis 7, Neo4j 5 (APOC)
CLAUDE.md                binding project memory, auto-loaded by Claude Code
ARCHITECTURE.md          condensed working reference
docs/                    this document and its predecessors
signals/                 indicators.py, scorer.py   (Stage 9)
tests/                   test_indicators.py, test_scorer.py, test_universe_leakage.py
tests/invariants/        adversarial invariant tests (run by CI)
requirements/base.txt    base dependencies
data/raw/                bhavcopy, delivery, fno_bhavcopy, announcements
data/processed/          corporate_actions.parquet, pit_features.parquet, checkpoints
```

Ingestion and pipeline scripts (bhavcopy and delivery ingestion, corporate-action parsing, universe build, cost model, momentum baseline, index, FII/DII, USDINR and repo-rate loaders) sit alongside a shared ingestion module that provides the HTTP session, retry, atomic writes and the manifest.

14.3 Bring-up steps

1. **Clone and create an environment.** Create a virtual environment with Python 3.12 and install `requirements/base.txt`.
2. **Create `.env`** from the template. Secrets are read only through `config.py`. Nothing is hardcoded.
3. **Start services.** Run `docker compose up -d`, then `docker compose ps` and confirm Postgres, Redis and Neo4j report *healthy*. Services use development-only default passwords; change them for any shared or hosted environment.
4. **Run the test suites.** `python -m pytest tests/invariants/ -v` runs the invariant suite. `python -m pytest` runs everything. A `pytest.ini` makes the `signals` package importable.
5. **Check data layout.** `data/raw/` and `data/processed/` exist and match Section 5.4. Ingestion is resumable: re-running skips completed dates.

6. **Run a stage.** Each stage has a passing check (Section 15) that is executed against real row counts before the next stage starts.

Service	Container	Port	Used from
PostgreSQL 16	`lynx_postgres`	5432	Point-in-time feature store (about Stage 12 onward)
Redis 7	`lynx_redis`	6379	Snapshot cache
Neo4j 5 Community (APOC)	`lynx_neo4j`	7474 (browser), 7687 (bolt)	Knowledge graph (Stage 19 onward)

14.4 Continuous integration

A GitHub Actions workflow runs on every push and pull request to the main branch. It installs the base requirements, runs the invariant suite, and then **fails loudly if no invariant tests were collected**. A green run with zero tests would look like safety without providing any, so "no tests ran" is treated as a failure.

14.5 Working conventions

- Loguru for logging, never `print()`.
- Tenacity for retries on any network or database call.
- Checkpoint and resume for every batch pipeline over the full universe: progress is written to `data/processed/*_checkpoint.json` after each symbol, so an interrupted run resumes.
- Pure computation stays free of database and network dependencies where practical, so it is unit-testable.
- `config.py` is the only place a path, URL or credential is read. A new module that needs one adds it there.
- Every risky component gets an invariant test written **before** the implementation is trusted.
- Long crawls run detached in the background and write checkpoints, because processes can be reaped under memory pressure.

14.6 Working with coding agents

Multiple AI tools may write the backend and the clients. Three habits keep this safe.

- **`CLAUDE.md` is binding.** It loads automatically in every session and carries the forbidden list. A request that conflicts with it is flagged, not silently followed.
- **One session per repository at a time.** Concurrent sessions in one repository caused a bulk file deletion and reaped background jobs earlier in the build. Sessions that must run in parallel work in separate branches or worktrees.
- **Show diffs before commits**, and keep the architecture document and `CLAUDE.md` in step with each other. When they disagree, the canonical document wins and the other is stale.

15. Build order and status

15.1 Build order

The order is 26 stages plus four new VANTA stages. Two orderings look wrong and are deliberate: the **cost model comes at Stage 6**, before features, because it gates the go/no-go at Stage 7; and the **walk-forward harness comes at Stage 13**, before the model it judges. Both make the project fail early rather than late. No stage starts before the previous stage's check passes against real row counts.

#	Stage	Depends on	Passing check
1	Environment, config, canonical paths		CUDA available, config loads, no missing variables, one database path fixed
2	Repo scaffold, CI, invariant test skeleton	1	Compose comes up; the empty suite passes in CI
3	Bhavcopy ingestion and backfill	2	Row counts match the ticker times trading-day product. The true start of delivery data is measured and documented.
4	Corporate action engine	3	No adjusted return outside its band without an action record
5	Universe and point-in-time constituents	3, 4	Per-year membership lists; delisted names kept with terminal values
6	Transaction cost model	3	Round-trip cost computable for any security on any date, delivery and intraday
7	Momentum baseline through costs: GO/NO-GO	4, 5, 6	Positive net return means a real edge to pursue; negative means widen the horizon or tighten the universe, not more model capacity
8	Index and macro pipeline	3	Nifty, Bank Nifty, India VIX, repo, CPI, USDINR populated
9	Indicators and deterministic scorer	4	Indicator count matches OHLCV count; scorer returns a full audit dictionary for a sample ticker
10	India block: delivery, F&O, flows, pledging, circuits	3, 8	Delivery validated against the raw file; missingness masked, never zero-filled
11	Override gates and circuit/ban filters	10	Gates fire correctly on known historical examples

#	Stage	Depends on	Passing check
12	Point-in-time feature store	9, 10, 11	A read as of date *t* returns only what was knowable before *t*
13	Walk-forward harness	12	A deliberately leaky configuration produces an implausible IC and is flagged. Adds label purging.
13A	**VANTA label simulator**	4, 5, 6, 9, 10	N1 to N4, N6 and N11 pass. Coverage report by year. 20 sampled trades reconcile with a hand calculation.
13B	**Filtered regime posterior** (minimal 3-state HMM, forward recursion only)	8, 10	N7 passes. Available from the first date all four inputs exist (India VIX starts 2014-05-14) plus warm-up.
13C	**VANTA selection gate**	13, 13A, 13B	N5, N8 to N10 and N13 pass. On synthetic exchangeable data, realised FDR is at most q.
13M	**Milestone M1: GO/NO-GO #2**	7, 13A, 13C (13B for the regime arm)	Pre-registered criteria in Section 11.5
14	Baselines on net labels, all through the same gate: A0, A1, **B1 CatBoost**	13M	Rank IC, net return per published signal and realised FDR, with seed and fold spread
15	LightGBM lambdarank (ablation A13)	14	Ranking objective isolated
16	Announcement ingestion and entity resolution	3	Announcements ticker-tagged, date range verified
17	GDELT BigQuery extraction and NSE entity mapping	16	Plausible volume per year; entity-resolution precision measured on a sample
18	LLM extraction pipeline and LoRA fine-tune	16	Precision and recall per event type against 500 hand-labelled articles
19	Knowledge graph schema and writers	18	Entities and edges queryable; every fact has a `source_span`
20	KG feature emission and graph edge derivation	19	Every graph snapshot provably built from trailing data only
21	SSL pretraining with WFP	20	Loss converges. Per-fold temporal split asserted by test. N14 passes. A18 run on three folds, with time per fold recorded.

#	Stage	Depends on	Passing check
22	Encoders and **VANTA ranker**	21, 13A	Trains within 4 GB. N10 passes. Beats B1 on net return per published signal, or the result is reported as negative.
23	**Regime soft gating** over ranker heads	22, 13B	N7 holds end to end. A9 reported.
24	Gate integration on PRISM plus VANTA scores, with ATR trade levels	23, 13C	Realised FDR and coverage by regime, per fold
25	Ablations and era analysis	24	All results with seed and fold spread. A2, A3, A10, A14, A16, A17 and B1 answered. Core frozen after this.
26	Pipeline, API, clients, outcome loop, deployment (web and mobile)	25	Ten unattended runs; tier checks; public paper-trading record live from the immutable log; store submission

Build order and status (30 stages, dependency-ordered)

Phase 0 Foundation

- **1** Env, config
- **2** Scaffold, CI

Phase 1 Data foundation

- **3** Bhavcopy ingest
- **4** Corp actions
- **5** Universe PIT
- **6** Cost model
- **7** Momentum GO

Phase 2 Features

- **8** Index, macro
- **9** Indicators
- **10** India block
- **11** Override gates
- **12** PIT store

Phase 3 Harness, VANTA core, baselines

- **13** WF harness
- **13A** Label sim
- **13B** Filtered HMM
- **13C** Selection gate
- **13M** M1 GO/NO-GO
- **14** Baselines+B1
- **15** LightGBM

Phase 4 Language and knowledge

- **16** Announcements
- **17** GDELT
- **18** LLM extract
- **19** KG schema
- **20** KG features

Phase 5 PRISM + VANTA model

- **21** SSL + WFP
- **22** Encoders+ranker
- **23** Soft gating
- **24** Gate integr.

Phase 6 Science and production

- **25** Ablations
- **26** API, apps, ops

built and verified · built, rework / re-run pending · not started · VANTA stages (v5/v6)

Build order and status

15.2 Status snapshot (20 October 2026)

This snapshot comes from the most recent build reports (no new stage report since 9 October). It should be re-verified in the repository before it is quoted as fact.

Stage	State
1, 2	Built and verified. Docker services healthy.
3	Ingestion built. A cleanup is under way and drives re-runs of the stages that depend on it (4, 5, 8, 9, 10).
4, 5	Built. Re-run pending after the Stage 3 cleanup.
6	Built (date-versioned cost model).
7	Momentum baseline built; verdict was GO (about 4.6% net annualised long-short decile spread, monthly rebalance). The verdict is to be re-checked after the re-runs.
8, 9	Built. Re-run pending. Stage 9 indicators and scorer have unit tests.
10	Four of five acceptance checks pass. Open: the F&O feature block is still missing, and the FII/DII flows have no publication-timing evidence. That evidence cannot be backfilled, because the monitor that records it is not a daemon, so this item is documented as a known limitation rather than closed. One historical F&O day (2003-10-01) is excluded.
11 to 26, 13A to 13M	Not started. **No VANTA code is written until Stage 10 closes.** Component specifications and invariant tests (written before code) may be prepared in parallel in `docs/vanta/` and `tests/invariants/`.

15.3 Critical path

Stages 1 to 25 are the decision path and constitute a complete dissertation without the product. The path to the first real evidence is: finish Stage 10, build Stages 11 and 12, then Stage 13 and the VANTA stages 13A (NEL), 13B (filtered HMM) and 13C (RW-CS with FCC), then M1. M1 is the earliest point at which the new decision layer meets real NSE data, and it uses momentum scores, so it needs no new model.

15.4 Descope order

If time compresses, cut in this order. Everything above the line remains a complete and defensible dissertation.

1. Intraday horizon (already outside VANTA v1; marked future work).
2. GDELT historical news, falling back to announcements only, which carry most of the signal anyway.
3. Knowledge graph, keeping LLM extraction as flat features.
4. Multi-agent orchestration, falling back to a single extraction agent.

The line: Stages 1 to 25 alone are a complete dissertation. Below it the science itself is being cut. The mobile app is no longer droppable; it is a supervisor-mandated deliverable.

THE FAILURE MODE TO ACTUALLY EXPECT

Not running out of time. Spending week 20 tuning an agent prompt while the cost model is unwritten. The decision path is the critical path; the agent and knowledge layers make it presentable. In that order.

16. Risk register

Risk	Severity	Mitigation
No edge net of costs	Critical	Cost model at Stage 6 and momentum baseline at Stage 7 were built first. Gross-profitable and net-dead is the normal outcome. Fallbacks: longer horizon or tighter universe, never more model capacity.
Synthetic VANTA results do not transfer	High	Treated as design evidence only. M1 and ablations A2, A3 and A10 decide.
The FDR promise is read as a guarantee	High	Stated everywhere as a target. Realised FDR by regime is reported in backtests and in the live log.
Silent leakage	High	Invariant tests N1 to N14 and the catalogue in Section 12. An IC above 0.10 is treated as a bug report.
Pretraining does not transfer	High	A4 and A14 measure it directly. A negative result is publishable. The architecture degrades gracefully to B1.
Delivery data is shallow (from September 2019)	High	Measured at Stage 3. Per-fold availability and era analysis turn it into a finding.
Label window limited by circuit-list coverage	Medium	Rows before the first hit list are excluded from primary results. Inferred bands are not used.
Regime posterior starts in 2014	Medium	The regime-weighted gate is testable only from India VIX's start on disk. The age-only arm covers the earlier window.
Small effective calibration sample in rare regimes	Medium	Log the Kish effective sample size; fall back to age-only weights below the threshold.
60-session calibration lag	Medium	Outcomes mature late, so the 52-week window holds fewer matured rows. Consider a horizon-specific window.
Free cloud GPU limits	Medium	Checkpoint every epoch and design pretraining to resume. Budget one paid GPU week if needed.
Per-fold pretraining exceeds the GPU budget	High	From-scratch pretraining on every fold is several hundred GPU-hours. WFP (Section 8.1.1) warm-starts later folds. If WFP fails A18, headline PRISM results use fewer folds, and that is reported.

Risk	Severity	Mitigation
A named component adds nothing	Medium	Each component has a kill rule (Section 9.0). Dropping one is a reported result, not a failure of the project.
GDELT entity resolution is poor	Medium	Announcements are pre-tagged and carry most of the signal. GDELT degrades to a co-mention edge source.
Extraction quality is low	Medium	Measured at Stage 18 before the graph is built on top. Restrict to high-confidence event types if needed.
Scope overrun	High	Descope order in Section 15.4.
Index licensing	Medium	Sensex is BSE IP; Nifty and Bank Nifty are NSE Indices IP. Review display and redistribution terms before public launch.
Paid tiers against SEBI Research Analyst rules	Low likelihood, high impact	Build payments in test mode. Resolve registration before any store submission.
Coding-agent drift and concurrent-session accidents	Medium	`CLAUDE.md` forbidden list, one session per repository, diffs shown before commits.

17. Open decisions

1. **Horizon scope.** Confirm that VANTA v1 covers 5, 20 and 60 sessions and that intraday is future work (Section 2.5).
2. **Max-tier ungated list.** Recommendation: show it as a watchlist without entry, stop or target.
3. **Fallback threshold.** Kish effective sample size below about 200 triggers age-only weights. The number is a proposal.
4. **Horizon-specific calibration window** for the 60-session horizon.
5. **Incumbent calibration.** A separate calibration on incumbent net returns is a later option. VANTA v1 runs the gate on full-cost returns for all names, which is conservative for incumbents.
6. **SEBI Research Analyst registration** before store submission.
7. **Weighted-selection pruning** (Jin and Candès 2023b) as ablation A16(d).
8. **Number of PRISM folds.** All folds for baselines and M1. Whether the full PRISM model runs on every price fold or only on the folds where the regime inputs exist (2014 onward) depends on WFP's measured cost at Stage 21.
9. **Decay scale τ grid** for RW-CS: {13, 26, 52} weeks is proposed.

Appendix A. Data contracts

All tables are Parquet and append-only. Every row carries the version tags in A.4.

A.1 `vanta_labels`, partitioned by horizon and year

Column	Type	Notes
`symbol`	str	
`decision_date`	date	*t*. Features and ATR as of *t*'s close.
`horizon`	int	h in sessions: 5, 20 or 60
`label_version`	str	Encodes k_s, k_t, tie rule and lock rules
`cost_model_version`	str	Stage 6 rates version used
`entry_date`	date	*t+1*
`entry_price_raw`	float	Raw open. NaN if no fill.
`fill_status`	enum	FILLED, NO_FILL_UPPER_LOCK, NO_FILL_NO_TRADE, NO_DATA
`atr_pct`	float	ATR(14) divided by close at *t*
`k_s`, `k_t`	float	Stop and target multiples
`stop_price`, `target_price`	float	
`exit_date`, `exit_price`	date, float	
`exit_reason`	enum	TARGET, STOP, STOP_GAP, STOP_DELAYED_LOWER_LOCK, TIME, DELISTED
`lock_delay_sessions`	int	0 unless the exit was delayed by a lower lock
`lock_source`	enum	HIT_LIST or NONE. NONE rows are excluded from primary results.
`gross_return`	float	Adjusted-price return
`cost_entry`, `cost_exit`, `cost_impact`	float	Stored separately
`y_net`	float	Gross minus all costs (full round trip)
`label_end_date`	date	Equals `exit_date`. Used for purging.

Winsorisation and the gain g are computed at training time from fold-only statistics and are never stored as fixed columns.

A.2 `vanta_calibration`

Column	Type	Notes
`decision_date`	date	When the score was produced
`symbol`, `horizon`		
`score`	float	Out-of-sample score

Column	Type	Notes
`model_version`	str	Scores from different versions are not mixed
`fold_id`	str	
`is_oos`	bool	Must be true
`regime_posterior`	float[3]	**Filtered** π at `decision_date`
`hmm_version`	str	
`y_net`	float	Null until matured
`matured_at`	timestamp	Label end close plus publication lag. Null until matured.
`candidate_filter_version`	str	Same override-gate filter as the test candidates

Query rule at decision date T: `matured_at <= T` and `decision_date >= T - 52 weeks` and `model_version == current` and `candidate_filter_version == current` and `is_oos`. If the current model is newly trained, its out-of-fold scores for the past 52 weeks are written before the gate runs; otherwise the calibration rows would be scored by a different model than the test rows.

A.3 `vanta_gate_log`, immutable, one row per (decision date, candidate)

Per row: `decision_date`, `symbol`, `horizon`, `score`, `p_value`, `bh_threshold`, `in_bh_set`, `published`, `cap_binding`, `q`, `n_calibration`, `effective_n` (Kish effective sample size), and all version tags. Outcome fields (`y_net`, `lost`) are added once matured. Realised FDR and the pooled loss share are computed from this table only.

A.4 Version tags

Format: `{component}@{fold_id}:{train_end}:{git_sha}:{config_hash}`, for example `vanta-ranker@f07:2018-12-31:a1b2c3d:9f2e01`. Tagged components: `label_version`, `cost_model_version`, `encoder_version` (including the WFP parent checkpoint), `model_version` (the ranker), `hmm_version`, `gate_version` (q, cap, decay scale τ, distance) and `candidate_filter_version`. Every artifact is registered in MLflow with these tags, so a gate decision is reproducible from its log row alone.

Appendix B. Decision log

Decision	Why	Replaced
Bhavcopy, not yfinance, is the price backbone	yfinance drops delivery data and mis-adjusts Indian corporate actions	The earlier yfinance plan
Rolling, not expanding, walk-forward	Indian market structure broke repeatedly	Expanding window
Embargo equals the label horizon, per horizon	A 2-day embargo on a 3-day label leaks	A constant 2-day embargo
Cost model built before features (Stage 6)	The largest risk is no edge net of costs	Costs added late
FinBERT replaced by a structured extractor	Domain concepts and lossy scalar output	Per-ticker sentiment scalar
Qwen3-4B as the extractor	Fits the 4 GB local card for inference	Earlier 8B options
Mobile app is mandatory	Supervisor requirement at the Phase I presentation	Mobile as first descope item
Net executable labels	The product is judged on money after costs	Sector-demeaned forward-return rank
Volatility-normalised ListNet	Plain top-heavy ranking favours lottery stocks	LambdaLoss with NDCG
Conformal selection with BH at q = 0.3	The per-stock lower bound almost never cleared costs, and it controlled the wrong error	Per-stock conformal lower bound above cost
Soft regime gating by filtered posterior	Posterior-weighted rather than hard routing, with no learned network	Hard-routed regime heads
ATR-multiple published levels	The label must describe the published trade	Quantile-head levels (now display only)
Incumbency handled at accounting, not in labels	Avoids a circular, non-stationary target	Incumbency baked into labels
Filtered HMM pulled forward to Stage 13B	The gate needs it before the full model exists	Regime model at Stage 23
The learned “cash” item dropped	It never abstained in simulation; the gate does that job	A cash item inside the ranking
q described as a target, not a theorem	Exchangeability fails, weights are data-dependent, BH on weighted p-values is outside the theorems	Language implying a guarantee
VANTA defined as a named pipeline of four components (NEL, VN-ListNet, RW-CS, FCC)	The name covers the combination; each part keeps its original citation; each is tested and ablated separately	VANTA described loosely as “an algorithm”
WFP warm-started walk-forward pretraining	From-scratch pretraining on every fold did not fit the free GPU budget	One from-scratch pretraining run per fold
τ chosen on validation folds	A fixed 26-week decay was an untested constant	τ fixed at 26

Decision	Why	Replaced
“Gate-first inference” rejected	Filtering stocks out before PRISM would remove them as context nodes from the graph network and change every neighbour’s representation. The only saving left (SHAP and narration) already applies to published names only.	Proposed during the v6 review

Appendix C. Glossary

Term	Meaning
ATR	Average true range: a measure of a stock’s typical daily price movement. Stops and targets are set as multiples of it.
Bhavcopy	NSE’s end-of-day price and volume file
BH	Benjamini-Hochberg: a procedure that selects hypotheses while controlling the false-discovery rate
Circuit limit	The daily price band within which a stock may trade. A stock at its upper circuit can have buyers but no sellers.
Conformal prediction	A method that turns past errors into calibrated statements about new predictions
Delivery %	Share of traded volume that is actually delivered rather than squared off within the day
Embargo and purging	Removing training rows near the test window, so that labels cannot overlap it
FCC	Full-candidate calibration: the rule that the gate calibrates on every scored, matured candidate, not only published ones
FDR	False-discovery rate: the expected share of selections that are wrong. Here, the share of published signals that lose after cost.
Filtered and smoothed posterior	Filtered uses data up to now. Smoothed also uses the future and is forbidden in LYNX.
Kish effective sample size	$(\Sigma w)^2 / \Sigma w^2$: how many equally weighted samples a weighted set is worth
ListNet	A ranking loss defined over a whole ranked list
Net executable label (NEL)	The return of a simulated, publishable trade after all costs
PIT	Point-in-time: data restricted to what was knowable on a given date
R-GAT	Relational graph attention network
RW-CS	Regime-weighted conformal selection: the VANTA gate, with calibration weights from recency and the filtered regime posterior
Rank IC	Rank correlation between predicted and realised cross-sectional returns
SSL	Self-supervised learning: learning from the data itself without labels
STT	Securities Transaction Tax
VN-ListNet	ListNet with a volatility-normalised gain and soft regime gating: the VANTA ranker
WFP	Warm-started walk-forward pretraining: each fold’s encoder continues from the previous fold’s checkpoint
Walk-forward	Evaluation that trains on the past and tests on the next unseen period, repeatedly

Appendix D. References

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How each component relates to prior art. NEL extends triple-barrier labelling with Indian circuit-lock rules, stock-specific costs and an executable-trade definition. VN-ListNet applies ListNet and Plackett-Luce with a volatility-normalised gain. RW-CS applies conformal selection with time- and regime-weighted calibration in the spirit of Barber et al. FCC is standard conformal practice made into a tested rule. WFP applies continued pretraining to a leak-free walk-forward chain. The claim is limited to the combination, evaluated on NSE equities.